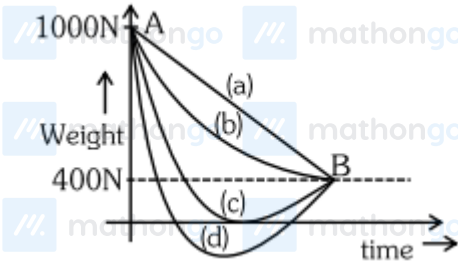


Q1 (20 July 2021 Shift 1)

A person whose mass is 100 kg travels from Earth to Mars in a spaceship. Neglect all other objects in sky and take acceleration due to gravity on the surface of the Earth and Mars as 10 m/s^2 and 4 m/s^2 respectively. Identify from the below figures, the curve that fits best for the weight of the passenger as a function of time.



(1) (c)

(2) (a)

(3) (d)

(4) (b)

Q2 (20 July 2021 Shift 2)

A satellite is launched into a circular orbit of radius R around earth, while a second satellite is launched into a circular orbit of radius $1.02R$. The percentage difference in the time periods of the two satellites is :

(1) 1.5

(2) 2.0

(3) 0.7

(4) 3.0

Q3 (20 July 2021 Shift 2)

Consider a binary star system of star A and star B with masses m_A and m_B revolving in a circular orbit of radii r_A and r_B , respectively. If T_A and T_B are the time period of star A and star B, respectively, then :

$$(1) \frac{T_A}{T_B} = \left(\frac{r_A}{r_B} \right)^{\frac{3}{2}}$$

Questions with Answer Keys

MathonGo

$$(2) T_A = T_B$$

$$(3) T_A > T_B \text{ (if } m_A > m_B \text{)}$$

$$(4) T_A > T_B \text{ (if } r_A > r_B \text{)}$$

Q4 (22 July 2021 Shift 1)

A body is projected vertically upwards from the surface of earth with a velocity sufficient enough

to carry it to infinity. The time taken by it to reach height h is

S.

$$(1) \sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

$$(2) \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

$$(3) \frac{1}{3} \sqrt{\frac{R_e}{2g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

$$(4) \frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

Q5 (25 July 2021 Shift 1)

The minimum and maximum distances of a planet revolving around the Sun are x_1 and x_2 . If the minimum speed of the planet on its trajectory is v_0 then its maximum speed will be:

$$(1) \frac{v_0 x_1^2}{x_2^2}$$

$$(2) \frac{v_0 x_2^2}{x_1^2}$$

$$(3) \frac{v_0 x_1}{x_2}$$

$$(4) \frac{v_0 x_2}{x_1}$$

Q6 (25 July 2021 Shift 2)

Consider a planet in some solar system which has a mass double the mass of earth and density equal to the average density of earth. If the weight of an object on earth is W , the weight of the same object on that planet

Questions with Answer Keys

MathonGo

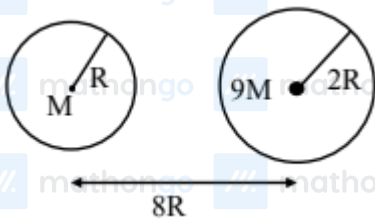
will be :

- (1) $2W$
- (2) W
- (3) $2^{\frac{1}{3}}W$
- (4) $\sqrt{2}W$

Q7 (27 July 2021 Shift 1)

Suppose two planets (spherical in shape) of radii R and $2R$, but mass M and $9M$ respectively have a centre to centre separation $8R$ as shown in the figure. A satellite of mass ' m ' is projected from the surface of the planet of mass ' M ' directly towards the centre of the second planet. The minimum speed ' v ' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7} \frac{GM}{R}}$ then the value of ' a ' is ____

[Given : The two planets are fixed in their position]



Q8 (27 July 2021 Shift 2)

Two identical particles of mass 1 kg each go round a circle of radius R , under the action of their mutual gravitational attraction. The angular speed of each particle is :

- (1) $\sqrt{\frac{G}{2R^3}}$
- (2) $\frac{1}{2} \sqrt{\frac{G}{R^3}}$
- (3) $\frac{1}{2R} \sqrt{\frac{1}{G}}$
- (4) $\sqrt{\frac{2G}{R^3}}$

Q9 (27 July 2021 Shift 2)

Questions with Answer Keys

MathonGo

The planet Mars has two moons, if one of them has

a period 7 hours, 30 minutes and an orbital radius of 9.0×10^3 km. Find the mass of Mars.

$$\left\{ \text{Given } \frac{4\pi^2}{G} = 6 \times 10^{11} \text{ N}^{-1} \text{ m}^{-2} \text{ kg}^2 \right\}$$

(1) 5.96×10^{19} kg

(2) 3.25×10^{21} kg

(3) 7.02×10^{25} kg

(4) 6.00×10^{23} kg

Answer Key**Q1 (1)****Q2 (4)****Q3 (2)****Q4 (4)****Q5 (4)****Q6 (3)****Q7 (4)****Q8 (2)****Q9 (4)**

Hints and Solutions

MathonGo

Q1

At neutral point $g = 0$ so graph (C) is correct

Hence option (1).

Q2

$$T^2 \propto R^3$$

$$T = kR^{3/2}$$

$$\frac{dT}{T} = \frac{3}{2} \frac{dR}{R}$$

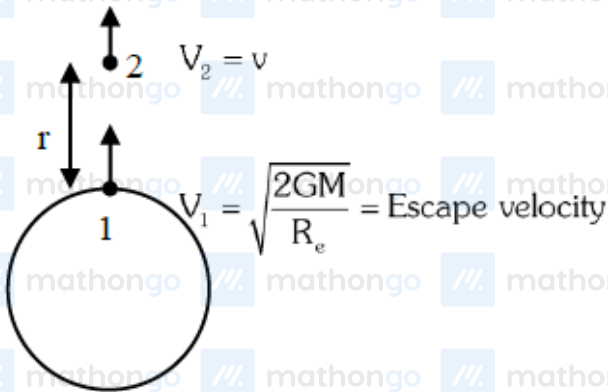
$$= \frac{3}{2} \times 0.02 = 0.03$$

$$\% \text{ Change} = 3\%$$

Q3

$$T_A = T_B \text{ (since } \omega_A = \omega_B \text{)}$$

Q4



Hints and Solutions

MathonGo

Applying energy conservation from (1) to (2)

$$\frac{1}{2} m \cdot \left(\frac{2GM}{R_e} \right) - \frac{GMm}{R_e} = \frac{1}{2} mv^2 - \frac{GMm}{R+r}$$

$$\Rightarrow \frac{1}{2} mv^2 = \frac{GMm}{R+r}$$

$$\Rightarrow v = \sqrt{\frac{2GM}{R+r}} = \frac{dr}{dt}$$

$$\Rightarrow \sqrt{2GM} \int_0^t dt = \int_{R_e}^{R_e+h} (\sqrt{R+r}) dr$$

$$\sqrt{2GM} \cdot t = \frac{2}{3} \left[(R+r)^{3/2} \right]_{R_e}^{R_e+h}$$

$$t = \frac{2}{3} \sqrt{\frac{R_e^3}{2GM}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

$$\frac{GM}{R_e^2} = g$$

$$t = \frac{1}{3} \sqrt{\frac{2R_e}{g}} \left[\left(1 + \frac{h}{R_e} \right)^{3/2} - 1 \right]$$

Q5

Angular momentum conservation equation $v_0 x_2 = v_1 x_1$

$$v_1 = \frac{v_0 x_2}{x_1}$$

Q6

Density is same $M = \frac{4}{3} \pi R^3 \rho$, $2m = \frac{4}{3} \pi R'^3 \rho$

$$R' = 2^{1/3} R$$

$$\omega = \frac{GMm}{R^2}$$

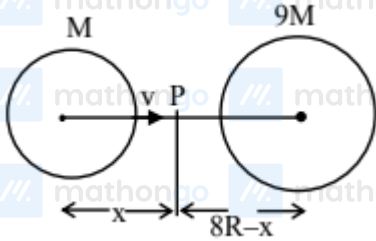
$$\omega_2 = \frac{G2Mm}{R'^2}$$

$$\omega_2 = 2^{1/3} \omega$$

Q7

Hints and Solutions

MathonGo



Acceleration due to gravity will be zero at P therefore,

$$\frac{GM}{x^2} = \frac{G9M}{(8R-x)^2}$$

$$8R - x = 3x$$

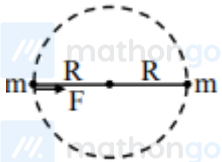
$$x = 2R$$

Apply conservation of energy and consider velocity at P is zero.

$$\frac{1}{2}mv^2 - \frac{GMm}{R} - \frac{G9Mm}{7R} = 0 - \frac{GMm}{2R} - \frac{G9Mm}{6R}$$

$$\therefore V = \sqrt{\frac{4}{7} \frac{GM}{R}}$$

Q8



$$F = \frac{Gm^2}{(2R)^2} = mR\omega^2$$

$$\omega = \frac{1}{2} \sqrt{\frac{G}{R^3}}$$

Q9

Option D is correct

$$T^2 = \frac{4\pi^2}{GM} \cdot r^3$$

$$M = \frac{4\pi^2}{G} \cdot \frac{r^3}{T^2}$$

by putting values

$$M = 6 \times 10^{23}$$